

SUMMER A TERM 2026 · 5 CREDITS · GEOGRAPHY 495

GeoAI

地理空间人工智能

A hands-on, critical introduction to machine learning, deep learning, large language models, and vibe coding for geospatial research.

GEOG 495

Geospatial Artificial Intelligence

Prof. Bo Zhao

Asynchronous online

GEOG 495

GEOSPATIAL

ARTIFICIAL

INTELLIGENCE

Build maps, models, prompts, and critical questions about how AI sees the world.

Learn GeoAI by doing

Use Google Colab, GitHub, Gemini, and open-source geospatial workflows.

From models to maps

Machine learning, CNNs, YOLO, SAM, LLM geocoding, and prompt-to-map experiments.

Vibe coding included

Use AI as a coding partner to prototype spatial analysis, debug notebooks, and explain results.

Critical, not just technical

Study hallucination, bias, surveillance, spatial justice, and the environmental cost of AI.

Four-Week Learning Arc

W1

Machine Learning for Spatial Data

Predict housing patterns, downscale urban heat, and connect models to geographic questions.

W2

Place, Space, and Deep Learning

Work with computer vision, object detection, and segmentation in socially sensitive contexts.

W3

LLMs for Mapping and Reasoning

Why Take It?

GeoAI is becoming a core skill for students who want to combine GIS, data science, public scholarship, and AI-assisted research. In this course, you will not only run models; you will learn how to ask better spatial questions, build reproducible projects, and use vibe coding responsibly as part of a research workflow.